Distributed Unsupervised Representation Learning for Remote Sensing Image Classification

Shaofan Li, Mingjun Dai, Yanglong Sun, and Yongfeng Suo

***Abstract*—With the development of in-orbit satellite hardware technology, distributed model training and updating for in-orbit satellites has become a promising trend. However, performance significantly degrades due to heterogeneous data distribution and the lack of high-quality labeled data across satellites. To address these issues, we propose a distributed unsupervised representation learning method called progressive uniformity (DURL-PU) from a deep clustering perspective. DURL-PU explores instance-level, cluster-level, and structure-level representations to progressively enhance uniformity, thereby mitigating dimensional collapse caused by data heterogeneity. Specifically, we first perform instance-level contrastive learning to distinguish between positive and negative samples. Second, we introduce a cluster-level loss to improve within-cluster compactness and between-cluster dispersion while incorporating high-level semantic information. Finally, we propose a deep clustering-based structural loss to enhance the consistency of parameter distribution among different satellites. Extensive experiments under various heterogeneous data scenarios demonstrate the effectiveness of DURL-PU.**

***Index Terms*—Deep clustering, distributed training, remote sensing (RS) image classification, unsupervised representation learning.**

I. Introduction

REMOTE sensing (RS) image classification, as a fundamental tool for RS interpretation [1], [2], plays a crucial role in environmental monitoring [3], [4], resource investigation, and urban planning [5], [6], [7].

Existing RS image classification methods typically adopt a centralized training structure, where RS images collected by each satellite are transmitted to a ground-based center for further processing. However, due to the large volume of RS image data, transmitting this data between satellites and the center consumes substantial bandwidth and resources. Additionally, this process introduces delays in providing feedback for tasks that require timely responses, such as disaster monitoring.

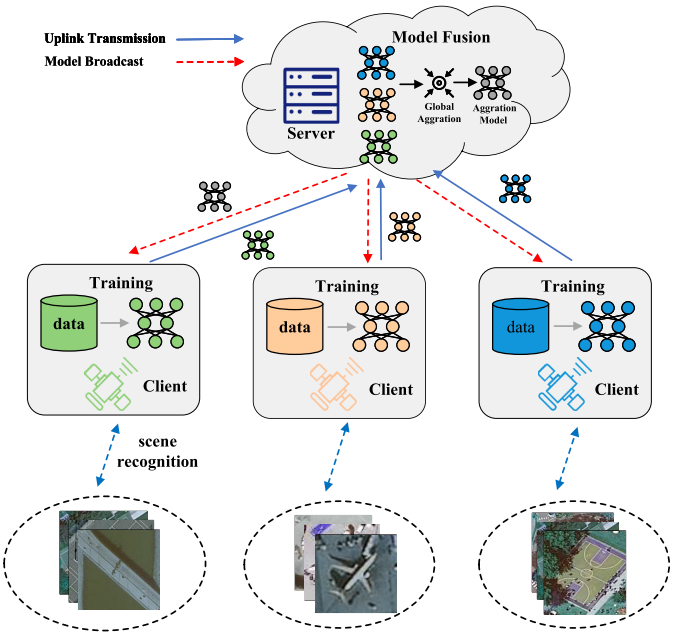


Fig. 1. Overall architecture of the distributed framework using FL.

With the rapid development of RS hardware and software technologies, the interpretation of models has gradually transitioned from ground-based processing to on-orbit processing [8], [9]. In on-orbit processing, some approaches utilize lightweight neural networks for training and deployment, which effectively improve response speeds [10], [11], [12]. However, these approaches rely on a single satellite, which is constrained by limited observation perspectives and computational capabilities, and thus cannot meet the high precision requirements for RS interpretation. The distributed architecture for on-orbit collaborative processing leverages data from multiple satellites, efficiently utilizing the computational and data resources dispersed across different satellites [13], [14], [15], [16], [17]. As shown in Fig. 1, each satellite (client) in a federated learning (FL) system keeps its raw data stored locally and performs model training and parameter updates on-device. Only the locally computed model updates (such as gradients or parameters) are transmitted to a central server, so that the raw data never leaves the satellite. As a result, FL provides a suitable framework for training a shared model across multiple satellites without the need for raw data

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transmission. FL ensures that data privacy and security are preserved by conducting both model training and parameter updates locally on each client device. The FL paradigm allows raw data to remain stored securely on distributed satellites, without being shared or centralized. Instead, only the locally computed model updates (e.g., gradients or parameters) are communicated to the server, significantly reducing the risk of data leakage.

However, the performance of a trivial application of FL degrades significantly due to data heterogeneity and label deficiency. On one hand, in terms of label deficiency, it is impractical to acquire a large number of precisely labeled samples for RS images, as it entails high labeling costs and expertise requirements. In general, only a small portion of RS images can be labeled, which greatly limits the performance of deep networks. On the other hand, regarding data heterogeneity, the images captured by different satellites exhibit variations, which potentially lead to label skew. For example, some satellites have a large number of images of airports, while others have few or none. This phenomenon is also referred to as statistical heterogeneity or non-independent and identically distributed (non-IID). Another heterogeneous scenario arises from variations in sensor configurations and data acquisition settings across different satellites, resulting in images of the same category presenting entirely distinct features, a phenomenon also referred to as feature distribution skew. Data heterogeneity is also common in transfer learning, where transfer learning addresses the distribution discrepancy between the source domain and the target domain through feature alignment, domain adaptation, or knowledge transfer, thereby enhancing the generalization ability of the model. Unlike transfer learning, FL involves multiple decentralized clients with significantly varied local data distributions. This leads to unique issues such as model aggregation difficulties, optimization instability, the need for balancing global collaboration with local personalization, and accuracy degradation caused by representation collapse, which this article aims to address.

There are some efforts devoted to addressing the data heterogeneity challenges, especially in skewed data distribution [18], [19], [20], [21], [22]. Li et al. [23] proposed a data synthesis method to generate features related to real samples, assisting in the training of local models and mitigating the impact of non-IID data. Diao et al. [24] proposed a simple and efficient method, FedConcat, which aggregates local knowledge by concatenating these local models based on the global model, achieving accuracy improvements in non-IID scenarios. Qu et al. [25] found that simply replacing the original convolutional network with a transformer architecture can effectively handle heterogeneous data and accelerate model convergence. However, the success of these methods primarily relies on supervised pre-training on ImageNet, which may be affected by domain differences when fine-tuning on RS data. This issue can be mitigated by pre-training using self-supervised learning on a centrally shared large-scale RS dataset. However, due to data silos, such centralized large-scale RS data is very rare in the real world; it is necessary to develop a self-supervised federated framework to jointly learn from all available unlabeled RS data across different satellites.

Label deficiency is a common challenge in RS image processing [26], [27], [28], [29], [30], which can limit the performance of deep learning models. These models require a large amount of labeled samples for accurate training. To address this issue, many semi-supervised or self-supervised methods have been proposed to enable models to learn knowledge from partially labeled or unlabeled data. However, these methods are generally designed for centralized learning scenarios and do not directly address the unique challenges in federated or distributed settings.

In this work, in order to conduct distributed collaborative training on unlabeled data from different satellites, we propose a distributed unsupervised representation learning framework from a deep clustering perspective. For the issue of label deficiency, we consider the model updates on different satellites as multiple training tasks in unsupervised representation learning. Unlike supervised learning, which heavily relies on label information, unsupervised representation learning mines the intrinsic features of the images without any labels and hence significantly reduces the disadvantage of label scarcity. Through this approach, our method enables the interaction of model parameters across different satellites to learn shared knowledge. While domain adaptation and few-shot learning are effective in addressing the scarcity of labeled data, they do not directly tackle the distinct issue of model collapse caused by data heterogeneity in FL scenarios. Our proposed method introduces progressive uniformity-based unsupervised representation learning, which enhances the consistency of model representations across clients while incorporating high-dimensional semantic information, effectively addressing this issue. Unlike existing methods that rely on domain adaptation or centralized feature alignment, our method leverages a novel deep clustering approach to enhance representation uniformity across clients. This innovation effectively improves the global model's generalization ability in federated settings.

Data heterogeneity is a major challenge in FL, as it can lead to representation collapse, where the global model's representations tend to fall into a low-dimensional space rather than the ambient space. The representation collapse of the global model is inherited from the collapse of local models. Therefore, our method explores the instance, cluster, and structure-level representation in the local model to progressively enhance the uniformity, which can mitigate dimensional collapse caused by data heterogeneity. Specifically, we first perform the instance-level contrastive learning to distinguish positive and negative samples. Second, we introduce cluster-level loss to enhance the within-cluster compactness and the between-cluster dispersion while incorporating high-level semantic information. Finally, we propose a deep clustering-based structural loss to enhance the consistency of parameter distribution among different satellites. Although both the cluster-level loss and the deep clustering-based structural loss have been studied in natural image classification, this work uniquely extends these strategies to the FL paradigm. To specifically address the issue of model collapse caused by data heterogeneity across decentralized clients, we have designed tailored algorithms

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that leverage clustering to enhance representation consistency and robustness across different clients. This novel application introduces a fresh perspective, effectively improving performance in non-IID FL scenarios. As shown in Fig. 2, our method effectively alleviates the issue of dimensional collapse caused by data heterogeneity. To the best of our knowledge, this is the first work to combine unsupervised representation learning with FL in the field of RS. The motivation for this work also stems from the unique challenges in remote scene image classification. Traditional remote scene image classification methods often rely on centralized training, requiring raw data to be transmitted to a central server, which raises concerns regarding privacy and communication overhead. While some studies [21], [31] have explored FL for scene image classification, they heavily depend on large-scale labeled datasets, which are costly and labor-intensive to obtain. Furthermore, they overlook a key characteristic of scene image classification data: high inter-class similarity and low intra-class variance, which complicates representation learning and increases classification confusion. The main contribution of this work is summarized below.

1) We propose a distributed unsupervised representation learning method called progressive uniformity (DURL-PU), a novel federated unsupervised image classification framework, designed for on-orbit training on unlabeled data from distributed clients in RS, alleviating the challenge of label deficiency. It also effectively mitigates the issue of dimensional collapse in model representations caused by data heterogeneity across clients in FL.
2) We introduce a progressive uniformity strategy to mitigate representation collapse caused by data heterogeneity in FL. It consists of the following.
   a) *Instance-level contrastive learning* to capture discriminative features without label supervision;
   b) *Cluster-level regularization* to enhance intra-class compactness and inter-class separability;
   c) *Structure-level alignment* to enforce consistent representation distributions across clients.
3) Experimental results show that our DURL-PU outperforms other state-of-the-art algorithms across various data distribution scenarios.

The rest of this article is organized as follows. In Section II, related works are presented. In Section III, the proposed DURL-PU is detailed. In Section IV, experimental result of proposed method is presented and analyzed. Finally, conclusion is drawn in Section V.

II. Related Works

First, we introduce distributed neural network frameworks, such as FL, and their advancements in RS. Second, we discuss recent developments in unsupervised representation learning.

A. Federated Learning

FL is a distributed machine learning framework that enables collaborative training of data from various clients without sharing clients' data with others. As shown in Fig. 1, the entire training process is conducted locally on the clients rather than uploading the raw data to a central server. This distributed architecture effectively utilizes dispersed data and computational resources while keeping data private at each client. McMahan et al. [32] first proposed an FL framework algorithm known as FedAvg. The core concept is to develop a global model by averaging the local model parameters from each client. However, data heterogeneity among clients can significantly degrade the performance of FL algorithms. FedProto [33] mitigated this problem by incorporating a proximal term during local training. Li et al. [34] proposed the FedProx, which updates the model using data prototypes instead of gradient information. This approach is suitable for scenarios involving both model and data heterogeneity, and it also enhances the training speed of the model. Karimireddy et al. [35] proposed the SCAFFOLD algorithm, which addresses model drift by incorporating a variance control term. This method is robust to data heterogeneity and can accelerate convergence. Wang et al. [36] proposed the FedNova, which addresses objective inconsistency through a normalized averaging method. This enables FedNova to perform effectively when dealing with non-IID data and device heterogeneity. Yang et al. [37] attempted to categorize the features in the data into performance-robust features and performance-sensitive features. By eliminating redundant information within the features, better results were achieved under data heterogeneity conditions.

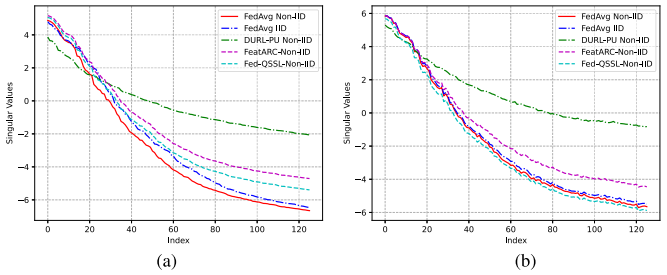


Fig. 2. Singular values of the covariance matrix of representations trained on (a) AID dataset and (b) FAIR1M dataset. The y-axis represents the logarithmic values of the singular values, while the x-axis denotes the index of the singular values. As we can see, with DURL-PU, the tail singular values are prevented from rapidly dropping to 0, indicating that it helps mitigate dimensional collapse and preserve the richness of the learned representations.

However, in the field of RS, research on FL remains in its early exploratory stage. Xu et al. [38] emphasize that FL is expected to become a fundamental technology for the future evolution of AI-driven RS. Wang et al. [31] proposed a distributed collaborative training method for RS image classification by using a sample proximity network to sample quality and category information, which is also the first algorithm to apply FL to RS tasks. Hu et al. [21] designed a personalized parameter aggregation strategy for few-shot RS image classification in an FL manner. Li et al. [39] formulate a novel multiparty personalized collaborative learning framework in terms of models and tasks, which can handle different models and tasks. Wang et al. [40] presented a personalized adaptive distillation module and multiscale feature matching network to get a more generalized and personalized model in the paradigm of multiparty collaborative learning.

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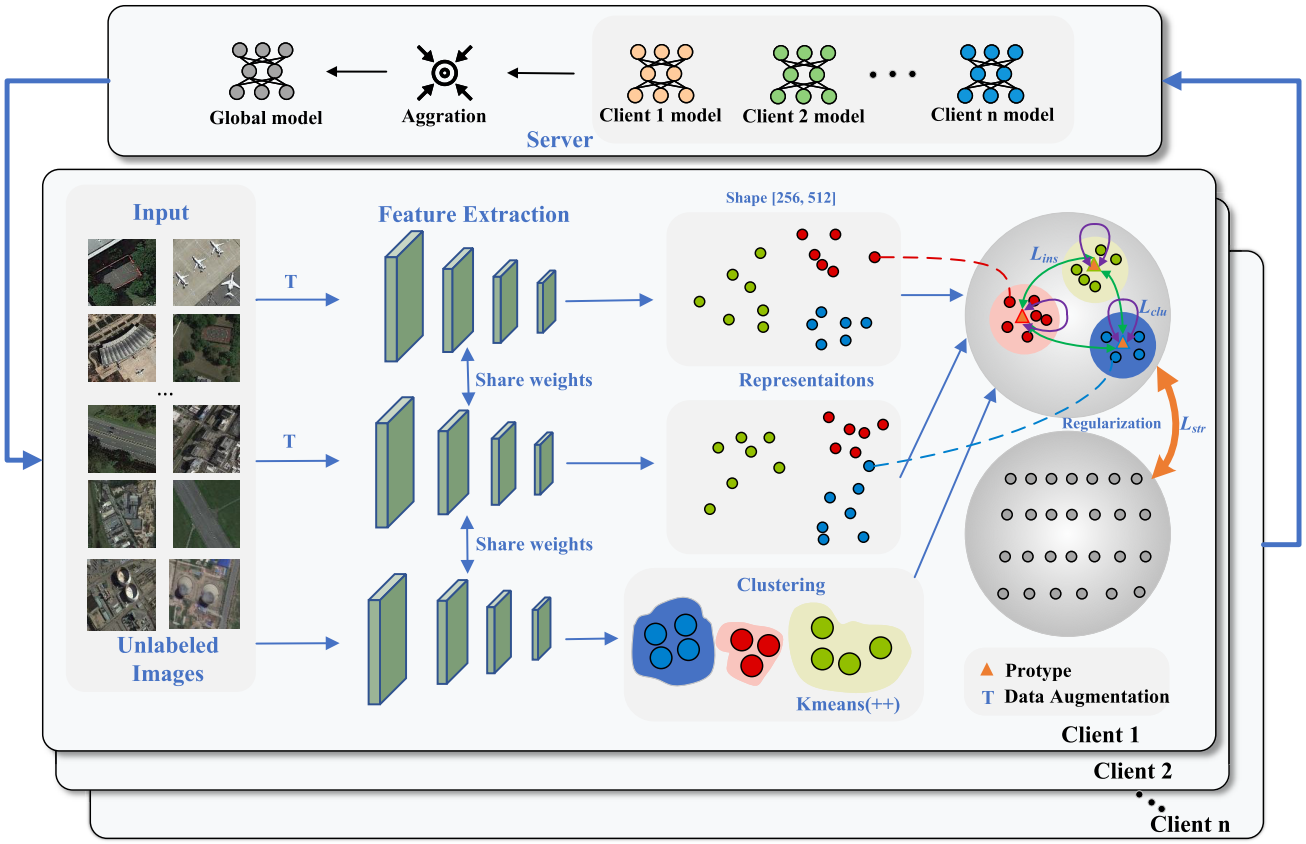


Fig. 3. Overall architecture of the DURL-PU.

Duna et al. [41] combined the intermediate features from multiple platforms to improve the accuracy of the object detection task. Li et al. [42] proposed a manifold-driven multimodality fusion framework to address the performance degradation caused by the non-IID characteristics of multimodal data. Büyüktaş et al. [43] conducted a systematic survey on FL algorithms for RS scene classification. They found that as the heterogeneity of training data increases, the performance of all FL algorithms experiences a significant decline.

B. Unsupervised Representation Learning

Unsupervised representation learning, also known as self-supervised learning, focuses on designing auxiliary tasks for unlabeled data to extract the inherent representational features of the data as supervisory information. Image-based self-supervised learning methods can be categorized into contrastive learning and generative approaches. This study belongs to the former category. Contrastive learning constructs positive and negative sample pairs through data augmentation to learn image differences, enabling the training of unlabeled data.

Zhao et al. [44] proposed an efficient multisample contrastive learning framework (EMSCNet) that constructs a dynamic dictionary to mine positive and negative sample pairs from the entire dataset. Xu et al. [45] developed a contrastive learning neural network (ACL-Net) incorporating an attention mechanism, which enhances the network's feature extraction capability by aligning and reinforcing target regions. Liu et al. [46] designed a multilevel feature alignment network to explore cross-model contrastive relationships and high-level semantic information. Li et al. [47] obtained more discriminative image knowledge by conducting contrastive learning on image information at different scales and granularities. Chen et al. [48] proposed a general-specific prototype contrastive learning (GSPCL) method to address the issues of inter-class similarity and intra-class variability in RS images. Zhang et al. [49] enhanced downstream detection generalization by unifying the structural encoding of representations through a combination of adversarial and self-supervised training. Zhou et al. [50] proposed an innovative contrastive learning framework for new class discovery. By introducing a positive sample selector, which enhances feature representation on unlabeled data by identifying and utilizing more positive samples from new classes. Gong et al. [51] proposed a dual-path aggregation attention network structure with quad-patch data augmentation to enhance the generalization of the extracted features. Liu et al. [52] proposed the scene-guided dual-branch network (SDNet), which leverages CAM technology to automatically distill knowledge from remote scenes that is closely related to salient objects. Additionally, it generates scene-enhanced saliency representations through a subnet, effectively improving the performance of saliency detection. Li et al. [53] proposed a multiform ensemble self-supervised learning framework to address the issue of low interclass separability in few-shot RS scene classification.

In summary, existing unsupervised representation learning methods for RS image classification typically rely on centralized training, which requires data to be stored on a single machine. However, due to the privacy concerns associated

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with RS data, it is difficult to collect all distributed RS data into a single node. Additionally, this approach consumes significant communication resources, making it unsuitable for tasks that require timely responses, such as disaster prediction. Therefore, developing a generalized model that can work with unlabeled data while ensuring data privacy is a highly valuable research direction in the field of RS.

III. Proposed Method

We propose a distributed unsupervised representation learning framework for RS image classification. As shown in Fig. 3, we formulate a clustering-based prototype loss to address the issue of representation collapse, a common problem in FL caused by data heterogeneity. Additionally, we enhance the uniformity of representations by computing the Wasserstein distance between the multivariate distribution of the clusters and a uniform Gaussian spherical distribution. This approach helps mitigate the accuracy degradation caused by inconsistent representation space distributions across different clients. Our method enables the network to handle data heterogeneity more effectively in real-world distributed RS scenarios.

A. Overall Workflow

The training process follows a federated optimization paradigm. At the beginning of the first communication round, the server initializes the global model and broadcasts it to all clients to ensure a consistent starting point. Regarding the relationships between different components, we explain as follows. First, we compute the first component, instance contrastive learning, which captures distinctions between individual samples. Second, we perform clustering on different samples to obtain class-specific clusters, leveraging cluster prototypes to enhance within-cluster compactness and between-cluster dispersion. Finally, the third component computes the mean and variance of each cluster and aligns the representation distribution with the unified spherical Gaussian distribution through the Wasserstein distance. Each client then updates its local model using a progressive three-stage learning process.

1) *Instance-level contrastive learning:* Each client first performs contrastive learning at the instance level by distinguishing positive and negative samples. Augmented views of the same sample are treated as positives, while others are considered negatives. Specifically, in the first part, Instance-Level Contrastive Learning, the input to the neural network is a batch of unlabeled samples. Each sample is augmented into two different views, which are then separately fed into the same neural network backbone (ResNet-18 without the final classification layer, denoted as Feature Extraction in Fig. 3). As a result, we obtain two sets of embedding vectors with dimensions [batch size, feature dimension], corresponding to the two augmented views. These are referred to as the representations. In this representation space, the horizontal axis corresponds to different samples, while the vertical axis corresponds to the feature dimensions of each sample. These embeddings are then passed through a multilayer perceptron (MLP) and a normalization layer to be projected onto a hyperspherical space in preparation for Instance-Level Contrastive Learning. We then compute the $\mathcal{L}_{\text{in}}$ loss value, as defined in (3), and the output of this component is a scalar loss used to update the model.
2) *Cluster-level representation learning:* To further refine feature representations, a cluster-level loss is introduced. This encourages within-cluster compactness and between-cluster dispersion, incorporating high-level semantic information derived from deep clustering. Specifically, in this part, Cluster-Level Representation Learning, we first feed the original, unaugmented image samples into the encoder to obtain embedding vectors with dimensions [batch size, feature dimension]. The K-means clustering algorithm is then applied to group the embeddings into different clusters, and the prototype information is extracted based on the cluster assignments. These prototypes are used to compute the loss defined in (4). The output of this module is the cluster-level loss $\mathcal{L}_{\text{clu}}$.
3) *Structure-level consistency optimization:* Finally, a deep clustering structural loss is applied to enhance the consistency of parameter distributions across different clients. This step mitigates the adverse effects of data heterogeneity by aligning the local model representations with the global model distribution. Specifically, in the final stage, the input consists of the cluster information obtained from the previous K-means clustering. We compute the mean and covariance matrix for each cluster, and then derive the overall mean and covariance of the representation distribution. Based on these statistics, we calculate the Wasserstein distance between the current distribution and the spherical Gaussian Distribution using (13), which serves as the structural loss $\mathcal{L}_{\text{str}}$. The output of this stage is the final loss value $\mathcal{L}_{\text{str}}$.

After the local updates guided by the three loss functions are completed through backpropagation, each client sends its locally updated model to the server. The server aggregates these models to obtain the global model, which is then distributed to all clients for the next communication round. This iterative process continues for a predefined number of communication rounds. When all training rounds are completed, we obtain the final trained global model (encoder). The overall training process of our DURL-PU is presented in Algorithm 1.

B. Object Function of FL

The overall FL framework employed in this article is depicted in Fig. 1. This framework consists of a central server and N clients, each client with its own local private dataset is denoted by $\mathcal{D}_k$. Both the central server and the clients are satellites, with each client connected to the central server. The central server is responsible for aggregating the models trained independently by each client and redistributing the aggregated model back to the clients. FL aims to develop a shared global model without requiring data sharing among

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Algorithm 1 Training Algorithm

Input: Dataset $D = \{x\}$, T represents the number of communication rounds, E is the number of client training epochs, initialization of global model $\{\theta^t\}$ and local models $\{\theta_n^t\}_{i\in[N]}$

Server executes:
for $t = 0,..., T - 1$ **do**
  Broadcast the global model w to each client
  for each client $n = 1,..., N$ in parallel **do**
    Update local model: $\theta_n^t \leftarrow \text{LocalUpdate}(E, \theta^t)$
  end for
  $\theta^{t+1} \leftarrow \frac{1}{N}\sum_{n=1}^{N}\theta_n^t$
end for
Return θ^T
LocalUpdate: (E, θ^t)
repeat
  E-step: Update $\{p(k|x)\}$ of the D_k dataset samples using k-means clustering algorithm.
  M-step:
  repeat
    for x_i in each local batch B **do**
      Randomly augment the x_i to obtain x^- and x^+
      Calculate the cluster centers using Eq (5).
      Calculate $\mathcal{L}_{\text{in}}$ using Eq (3)
      Calculate $\mathcal{L}_{\text{clu}}$ using Eq (4)
      Calculate $\mathcal{L}_{\text{str}}$ using Eq (13)
    end for
    Compute overall loss $\mathcal{L}$ using Eq (14)
    Update local model: $\theta_n^t \leftarrow \theta^t - \gamma\nabla\mathcal{L}(\theta^t)$
  until one epoch finished
until reaching E epochs

clients. The objective function for the distributed optimization of this global model is as follows:

$$\min_{\theta} f(\theta) = \frac{1}{N}\sum_{k=1}^{N} f_k(\theta; \mathcal{D}_k) \tag{1}$$

where $f(\theta)$ and $f_k(\theta)$ represent the server loss function and the kth client loss function, respectively, θ denotes the model parameters on each client. In FL, the global model is obtained by averaging the parameters of all local models. Specifically, the objective function for the local model is as follows:

$$\min_{\theta} f_k(\theta) = \frac{1}{N_k}\sum_{j=1}^{N_k} \ell_k\left(w; \mathcal{D}_{k,j}\right) \tag{2}$$

where N_k is the number of samples in $\mathcal{D}_k$, and $\mathcal{D}_{k,j}$ is the jth sample in the dataset $\mathcal{D}_k$. $\ell_k(\cdot;\cdot)$ is the self-supervised loss function of the kth client, which will be introduced later in this article.

C. Instance Contrastive Learning

Instance contrastive learning captures the similarities and differences between instance samples from different perspectives through InfoNCE loss [54]. Specifically, we start with an instance-level RS image x, which we augment randomly to obtain x^+. Additionally, we have M negative samples from the dataset x_i^-. The objective of this contrastive learning approach is to learn an embedding function $f(x)$ that maps data points to a unit hypersphere space. This mapping ensures that similar data points (positive pairs) are pulled closer together in the embedding space, while dissimilar points (negative pairs) are pushed apart. The contrastive loss function $\mathcal{L}_{\text{in}}$ promotes these relationships by maximizing the similarity of positive pairs and minimizing the similarity of negative pairs, based on the dot product between their embeddings. This process inherently increases the mutual information between similar samples, creates a clear density ratio distinguishing between positive and negative pairs, and reinforces the notion of similarity within the embedding space. Typically, the contrastive loss function for an instance sample can be defined as follows:

$$\mathcal{L}_{\text{in}} = -\log \frac{\exp\left(\frac{f(x)^T f(x^+)}{\tau}\right)}{\exp\left(\frac{f(x)^T f(x^+)}{\tau}\right) + \sum_{i=1}^{M}\exp\left(\frac{f(x)^T f(x_i^-)}{\tau}\right)} \tag{3}$$

where τ represents the temperature coefficient, and $f(x)$ is the embedding of x in a $L2$-normalized unit hypersphere. In FL, since data remains inaccessible across clients, this process is handled locally on each client. Specifically, for a given batch of data, each data instance undergoes two distinct augmentations to create two views of the same instance, which are treated as positive samples. For the same batch, all other augmented instances are considered negative samples. For example, if the batch size is n, there will be $2n$ views (two augmentations for each instance), resulting in 1 positive pair and $2(n-1)$ negative pairs for each instance. In summary, the criteria for determining positive and negative samples are as follows: For a given sample, its two different augmented views are considered a positive pair, while all other samples in the mini-batch (along with their augmented views) are treated as negative samples. This setup ensures that the positive and negative relationships are established entirely within each client without sharing raw data, preserving privacy while enabling contrastive learning effectively.

D. Contrastive Prototypes Uniform

Considering the issue of dimensional collapse caused by data heterogeneity, we propose a contrastive prototypes uniform (CPU) method to ensure that feature clusters with different semantics are distributed as uniformly as possible. This method also incorporates high-dimensional semantic information, compensating for the lack of semantic information in instance-level contrastive learning. As demonstrated in instance-level contrastive learning, this approach primarily focuses on bringing samples closer or pushing them apart, often from a microscopic perspective, which lacks a more global consideration. Li et al. [55] proposed PCL, which identifies prototype information through clustering, thereby bringing samples with the same high-dimensional semantic information closer together. However, although this method successfully reduces the distance between samples with similar semantics, it is not suitable for the heterogeneous FL environment. It tends to confine feature dimensions to a lower-dimensional space, leading to the problem of dimensional

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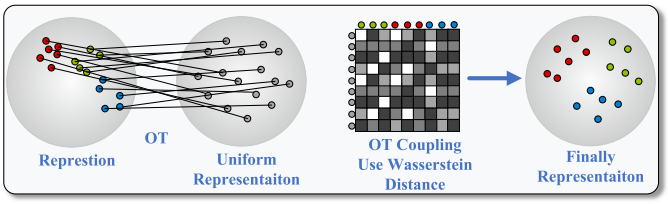


Fig. 4. Illustration of the structural representation distribution learning module. The core idea is to align the representation of each client distribution toward a uniform target distribution. To achieve this, we adopt optimal transport (OT) to model the transformation between distributions, where the Wasserstein distance serves as a specific metric to quantify and minimize the discrepancy between them.

collapse. The CPU method proposed in this article not only enhances intra-cluster compactness but also increases inter-cluster dispersion. The improvement in inter-cluster uniformity effectively mitigates the dimensional collapse phenomenon caused by data heterogeneity.

Specifically, we cluster the final features of the images into k prototypes, denoted as $\{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \ldots, \boldsymbol{\mu}_K\}$, among which there are r negative prototypes. Our proposed CPU is defined as follows:

$$\mathcal{L}_{\text{clu}} = \frac{1}{K}\sum_{k=1}^{K} -\log \sum_{j=1, j\neq k}^{K} \exp\left(\frac{\boldsymbol{\mu}_k^T \boldsymbol{\mu}_j}{\tau}\right) + \frac{1}{K}\sum_{k=1}^{K} \log \frac{\exp\left(\frac{f(x^+)^T \boldsymbol{\mu}_k}{\tau}\right)}{\sum_{j=1}^{r} \exp\left(\frac{f(x^+)^T \boldsymbol{\mu}_j}{\tau}\right)}. \tag{4}$$

The calculation method for the cluster center $\boldsymbol{\mu}_k$ is as follows:

$$\boldsymbol{\mu}_k = \frac{\sum_{x\in\mathcal{B}} p(k|x) f(x)}{\| \sum_{x\in\mathcal{B}} p(k|x) f(x) \|_2} \tag{5}$$

where $\mathcal{B}$ represents the batch size, and $p(k|x)$ is the posterior probability of cluster assignment. Given that the calculation of $p(k|x)$ is crucial for optimizing CPU performance, we employ the EM framework throughout the training process. Specifically, we use the K-means algorithm, iteratively performing clustering in the E-step and minimizing the objective function in the M-step of each epoch. The detailed algorithmic procedure can be found in Algorithm 1.

E. Structural Representation Distribution Learning

Enhancing the consistency of representation space distribution is challenging, especially when representations are clustered within a small region. For instance, in a heterogeneous data setting, the model's representations tend to be biased toward a subset of the global data manifold. As a result, there is no incentive for local models to spread their representations to a hyperplane with greater separability, which is more suitable for downstream fine-tuning tasks. To address this issue, we induce the clustering distribution structure of each local model's encoder to follow a Gaussian spherical distribution, thereby reducing discrepancies between client representations, as shown in Fig. 4.

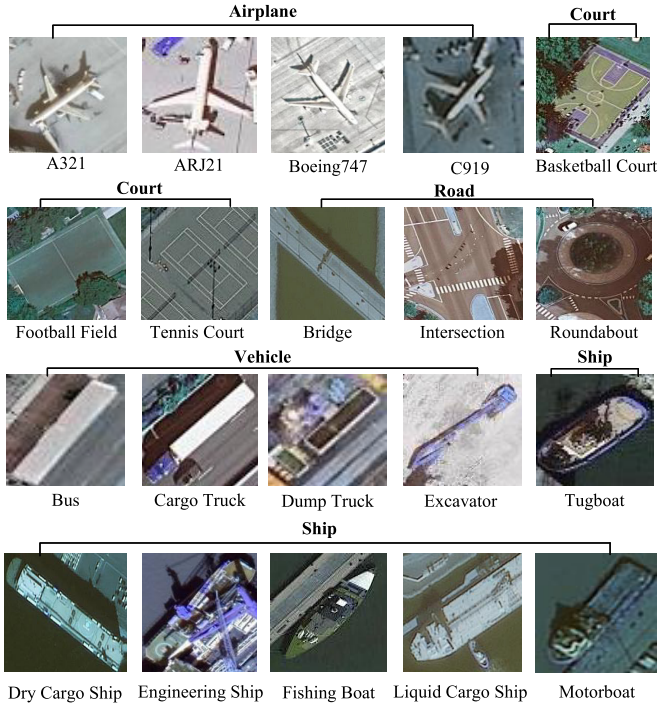


Fig. 5. Twenty representative object classes selected from FAIR1M.

The features learned by deep neural networks can be approximated as following a Gaussian mixture model distribution. In our approach, we assume that each cluster on the client side follows a Gaussian distribution. We gather local statistical information by calculating the mean and covariance for each cluster. Initially, we apply the K-means algorithm to each client to obtain c clusters $\{\boldsymbol{\mu}_{1,k}, \boldsymbol{\mu}_{2,k}, \ldots, \boldsymbol{\mu}_{c,k}\}$ in client k, $N_{c,k}$ is the number of sample in each cluster. The features of cluster c corresponding to client k are $\{\boldsymbol{z}_{c,k,1}, \ldots, \boldsymbol{z}_{c,k,j}\}$, where $\boldsymbol{z}_{c,k,j}$ is the feature of the jth sample in cluster c. The calculation of the mean $\boldsymbol{\mu}_{c,k}$ and covariance $\boldsymbol{\Sigma}_{c,k}$ for a single cluster is as follows:

$$\boldsymbol{\mu}_{c,k} = \frac{1}{N_{c,k}} \sum_{j=1}^{N_{c,k}} \boldsymbol{z}_{c,k,j} \tag{6}$$

$$\boldsymbol{\Sigma}_{c,k} = \frac{1}{N_{c,k}-1} \sum_{j=1}^{N_{c,k}} \left(\boldsymbol{z}_{c,k,j} - \boldsymbol{\mu}_{c,k}\right)\left(\boldsymbol{z}_{c,k,j} - \boldsymbol{\mu}_{c,k}\right)^T. \tag{7}$$

For a single client, the overall statistical distribution of $\mathcal{D}_k$ can be directly represented using the means $\boldsymbol{\mu}_{c,k}$ and covariances $\boldsymbol{\Sigma}_{c,k}$ of all clusters. The client's mean can be written as

$$\boldsymbol{\mu}_k = \frac{1}{N_c} \sum_{c=1}^{C} \sum_{j=1}^{N_{c,k}} \boldsymbol{z}_{c,k,j} = \sum_{k=1}^{C} \frac{N_{c,k}}{N_c} \boldsymbol{\mu}_{c,k}. \tag{8}$$

According to the definition in (7), the covariance can be expressed as follows:

$$\left(N_{c,k}-1\right)\boldsymbol{\Sigma}_{c,k} = \sum_{j=1}^{N_{c,k}} \boldsymbol{z}_{c,k,j}\boldsymbol{z}_{c,k,j}^T - N_{c,k} \cdot \boldsymbol{\mu}_{c,k}\boldsymbol{\mu}_{c,k}^T. \tag{9}$$

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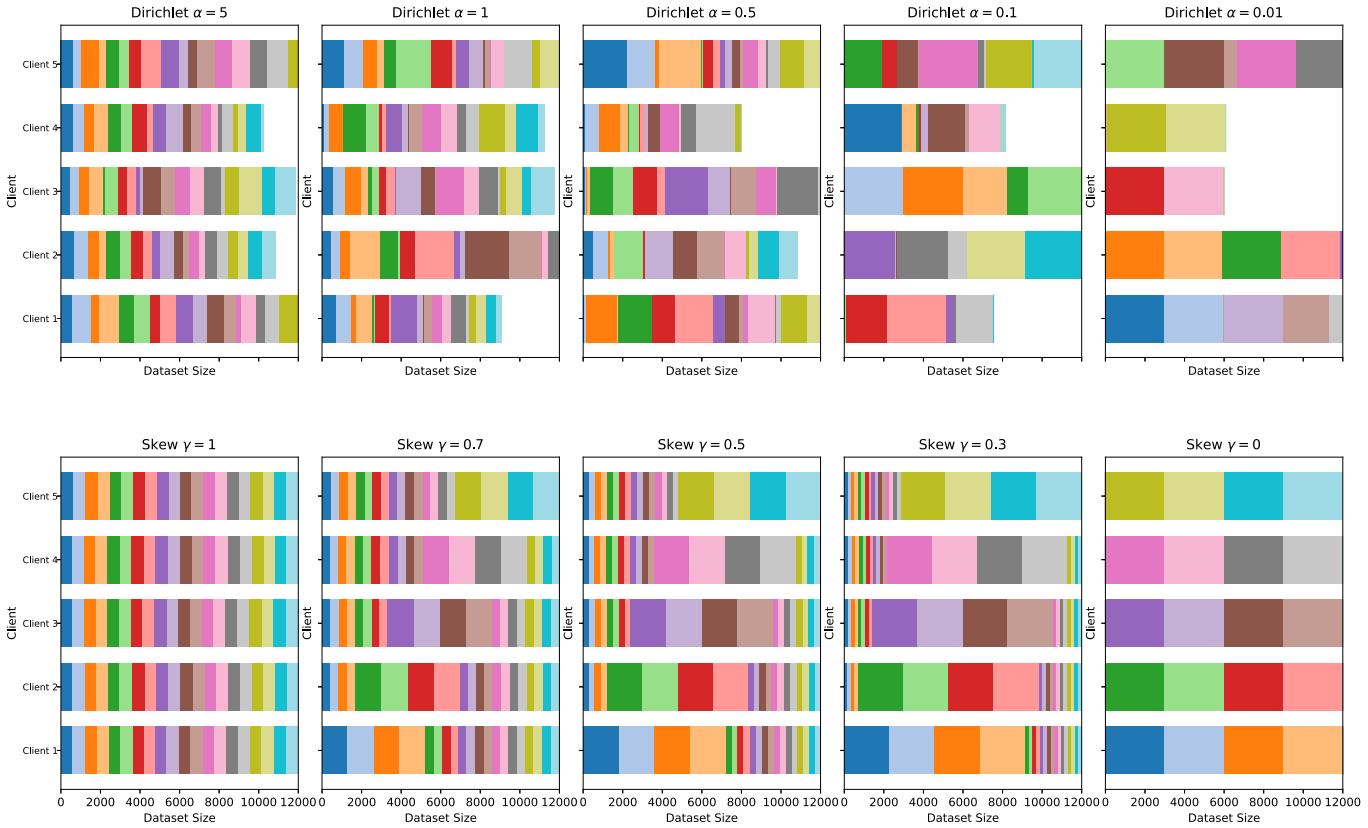


Fig. 6. Visualization of the data distribution shift for FAIR1M. Each vertical axis represents individual clients, while each horizontal axis denotes the data distribution across clients, with each color representing a class within FAIRM (20 classes totally). To investigate the impact of non-IIDness data on distributed learning, we use Dirichlet and skewness rates to generate diverse heterogeneous datasets. Our analysis shows that as we adjust these parameters, the datasets exhibit increasingly prominent non-IID characteristics from left to right.

When $N_{c,k} \geq 1$, the global covariance on the client side can be written as

$$\boldsymbol{\Sigma}_k = \frac{1}{N_c - 1} \sum_{c=1}^{C} \sum_{j=1}^{N_{c,k}} \boldsymbol{z}_{c,k,j} \boldsymbol{z}_{c,k,j}^T - \frac{N_c}{N_c - 1} \boldsymbol{\mu}_c \boldsymbol{\mu}_c^T. \tag{10}$$

After $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$ are obtained, we aim to align the feature distributions on each client with a unified spherical Gaussian distribution to mitigate issues such as accuracy degradation caused by distribution discrepancies across clients. The Wasserstein distance, also known as the earth mover's distance, is the minimal distance required to transfer distribution $P = N(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ to distribution $P = N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$. It is highly suitable for this scenario. The method for calculating this distance is as follows:

$$W(P, Q) = \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|^2 + B(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2) \tag{11}$$

where $W()$ is the Wasserstein distance function, $B()$ is referred to as the Bures–Wasserstein distance. It measures the distance between two covariance matrices $\boldsymbol{\Sigma}_1$ and $\boldsymbol{\Sigma}_2$. Specifically, it can be expressed as

$$B(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2) = \operatorname{tr}(\boldsymbol{\Sigma}_1) + \operatorname{tr}(\boldsymbol{\Sigma}_2) - 2\operatorname{tr}\left[\left(\boldsymbol{\Sigma}_1^{\frac{1}{2}} \boldsymbol{\Sigma}_2 \boldsymbol{\Sigma}_1^{\frac{1}{2}}\right)^{\frac{1}{2}}\right]. \tag{12}$$

Thus, we define our proposed structure learning as the Wasserstein distance between the previously computed cluster representation distribution $Z_1 = N(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ and the spherical Gaussian distribution $Z_2 = N(0, I)$

$$\mathcal{L}_{\text{str}} = \sqrt{\|\boldsymbol{\mu}_k\|_2^2 + 1 + \operatorname{tr}(\boldsymbol{\Sigma}_k) - \frac{2}{\sqrt{m}} \operatorname{tr}\left(\boldsymbol{\Sigma}_k^{\frac{1}{2}}\right)} \tag{13}$$

where m represents the batch size. We formulated the overall loss function for the joint training objective by incorporating representation learning at the instance level, cluster level, and structural level

$$\mathcal{L} = \mathcal{L}_{\text{in}} + \lambda_{\text{clu}} \mathcal{L}_{\text{clu}} + \lambda_{\text{str}} \mathcal{L}_{\text{str}} \tag{14}$$

where λ_{clu} and λ_{str} control the strength of each loss component, In this article, λ_{clu} is set to 1, and λ_{str} is set to 2. The selection of these loss weights plays a crucial role in determining the performance of the model. Specifically, λ_{str} is designed to enhance the consistency of representations across distributed client models, which is particularly important in FL scenarios with high data heterogeneity. By assigning a higher weight to λ_{str}, the method prioritizes reducing the divergence between local representations, thus improving the global model's robustness. Moreover, we carried out comprehensive experiments to analyze the impact of different weight settings in the subsequent experimental section. The results confirm that this particular configuration of loss weights leads to an optimal balance between enhancing representation uniformity and maintaining semantic distinctiveness,

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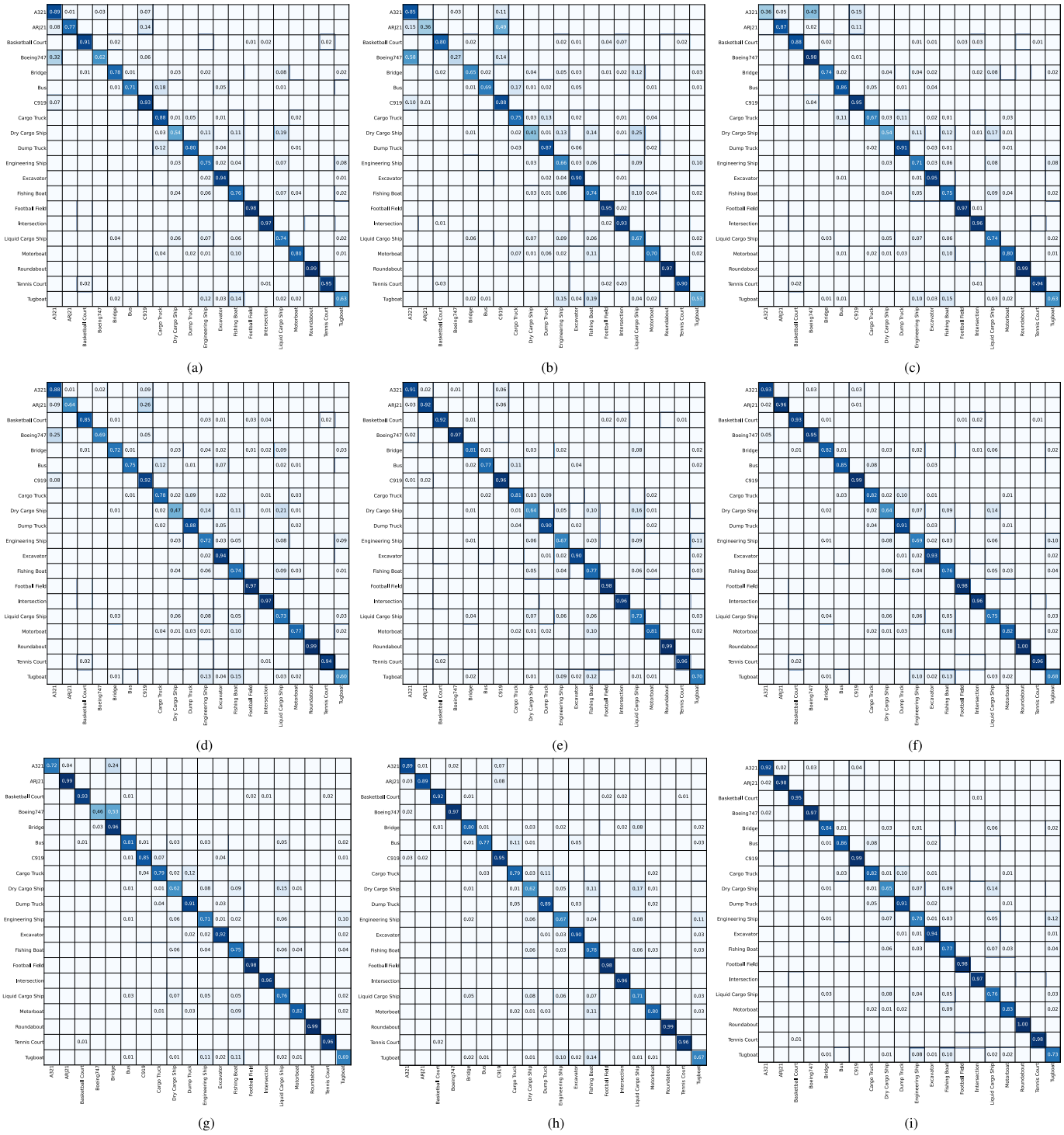


Fig. 7. Confusion matrix of different methods on the FAIR1M dataset. (a) FedSimCLR. (b) FedSimSiam. (c) FedBYOL. (d) FedMAE. (e) FedEMA. (f) FeatARC. (g) FedUSC. (h) Fed-QSSL. (i) DURL-PU.

ensuring the model's effectiveness under challenging non-IID conditions. The overall training process of our DURL-PU is presented in Algorithm 1. In the subsequent experimental section, we evaluate the effect of joint optimization by introducing corresponding loss functions at different stages for network optimization.

IV. Experiments

This section is divided into three main parts. The first part introduces the datasets used and our processing methods. The second part provides detailed information on the implementation settings of the experimental environment. The final part discusses the data validation methods and conducts an analysis of the experimental results.

A. Dataset

To validate the practical performance of the distributed unsupervised representation learning algorithm, we conducted a series of experiments on the RS image classification datasets,

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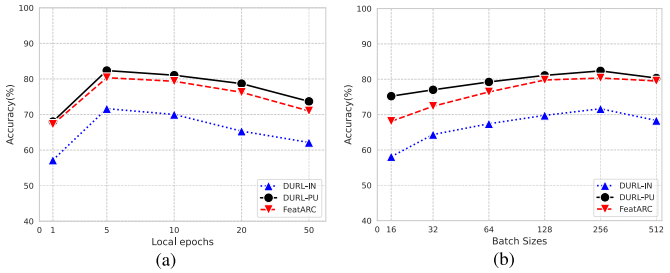


Fig. 8. Impact of local parameter of the client on algorithm performance. (a) Local epochs. (b) Batch sizes.

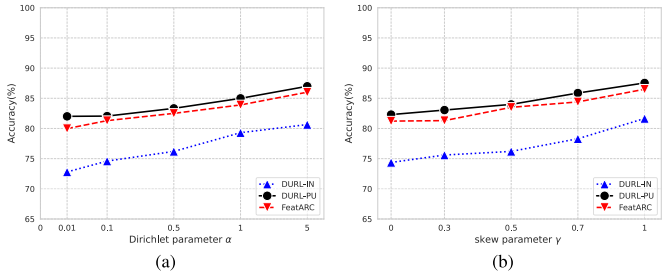


Fig. 9. Impact of data heterogeneity for algorithm performance. (a) Dirichlet parameter. (b) Skew parameter.

including FAIR1M [56], AID [57], UCM [58], and NWPU-RESISC45 [2]. FAIR1M is a new benchmark dataset designed for fine-grained object recognition in high-resolution RS images. It is currently the largest-scale dataset for fine-grained object recognition in high-resolution RS, with over one million instances from more than 15 000 images. The images in the FAIR1M dataset are collected from Gaofen satellites and the Google Earth platform, covering over 50 airports, ports, urban, and rural scenes worldwide. The AID dataset is an RS image dataset that contains scene images of 30 categories, with each category having approximately 220–420 images, totaling 10 000 images. Additional processing is necessary because of the vast number of instances in FAIR1M and the unique characteristics of its data. The UC Merced Land-Use Dataset (UCM) is a well-known benchmark dataset for RS scene classification. It consists of 21 land-use categories, such as agricultural, forest, and residential areas, with 100 images per category. The NWPU-RESISC45 dataset is a widely used benchmark for RS scene classification. It comprises 45 scene categories, including airport, farmland, and residential areas, with 700 images per category.

FAIR1M consists of five object categories: aircraft, vehicles, sports fields, roads, and ships. Each of these categories is further divided into several subcategories, resulting in a total of 37 fine-grained classes. It is important to note that the FAIR1M dataset contains a vast number of instances, exceeding one million, making the selection and division of categories a significant challenge. The object class annotations in the dataset are based on bounding box information, which introduces angle variations and dense annotation problems, potentially leading to the inclusion of irrelevant information in the object images. Additionally, some category images are very small objects with very low resolution, which can result in confusion during object classification. Therefore, it is necessary to adjust certain categories while ensuring a balanced distribution of data across the five major classes. Hence, careful division of the dataset is crucial.

Considering the aforementioned factors, we selected 20 representative classes from FAIR1M, as illustrated in Fig. 5. The chosen target classes have relatively high resolutions in FAIR1M, with clean backgrounds and minimal impact from dense annotations. Moreover, it is challenging to distinguish fine-grained subcategories within a single main category, particularly as these subcategories typically exhibit similar visual features, which can easily cause confusion for neural networks. For instance, distinguishing between various airplane models can only be achieved by examining the design of their wings, tail fins, and landing gear.

All images were cropped using sample bounding box information, with 5000 images per category. For categories with fewer than 5000 images, data augmentation techniques such as rotation, flipping, and color variation were used to supplement the samples. The dataset was split into training and testing sets in a 6:4 ratio. This approach produced an RS fine-grained dataset with 100 000 samples across 20 categories.

B. Implementation Details

The distribution of data is a key factor influencing the performance of distributed neural networks. Therefore, we first constructed a balanced data distribution to test the model's performance under ideal conditions. However, the data on distributed terminals in the real world often exhibit heterogeneous distributions. To simulate distributed clients in an FL environment on a single machine, the dataset is partitioned into multiple subsets, each representing the data of an individual client, with varying levels of heterogeneity achieved through specific partitioning strategies such as Dirichlet distributions. Each client independently trains a local model using its assigned subset, without accessing data from others, and communicates the updated model parameters or gradients to a central server. The server aggregates these updates to form a global model, which is then sent back to the clients for the next training round. This iterative process mimics real-world FL, where decentralized clients interact with a server. By leveraging simulation frameworks such as Pytorch, Flower, or TensorFlow, researchers can efficiently model client-server interactions and study the effects of data heterogeneity, communication constraints, and system scalability without requiring a physical distributed infrastructure. To simulate real-world scenarios, we further constructed two types of heterogeneous data distributions, including the Dirichlet distribution and the skewed distribution.

Non-IID data can be mathematically expressed as

$$P_i(X, Y) \neq P_j(X, Y), \quad \exists i \neq j \tag{15}$$

where the X represents the features, Y denotes the class labels, and $P_i(X, Y)$ refers to the data distribution on client i. which indicates that the joint data distribution differs across clients. A common type of non-IID data is label skew, where the class distributions vary across clients, which is also the type of heterogeneity considered in this article. This can be expressed as

$$P_i(Y) \neq P_j(Y), \quad \exists i \neq j. \tag{16}$$

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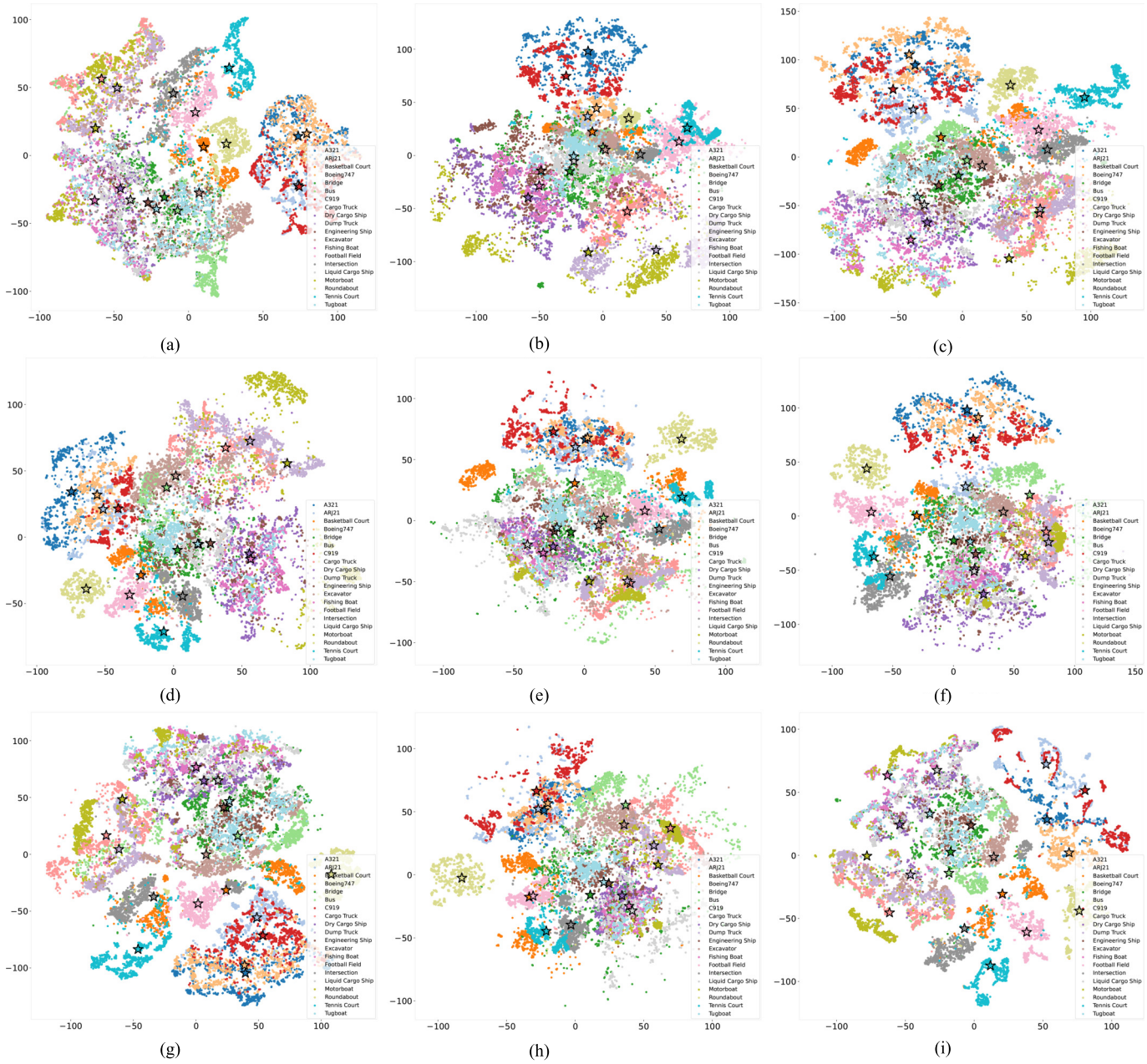


Fig. 10. T-SNE visualizations of features on unlabeled FAIRM test data with different methods, where stars represent the prototypes of different scene classes. (a) FedSimCLR. (b) FedSimSiam. (c) FedBYOL. (d) FedMAE. (e) FedEMA. (f) FeatARC. (g) FedUSC. (h) Fed-QSSL. (i) DURL-PU.

This indicates that certain classes are more prevalent on specific clients while being underrepresented or even missing on others. Such an imbalanced data distribution can affect the training of FL models, requiring specific strategies to mitigate its impact.

To generate heterogeneous data for distributed learning, we first employed the Dirichlet distribution [59]. Specifically, for each category, we controlled the data distribution across different clients using a parameter vector $\boldsymbol{\alpha}$. The probability density function of the Dirichlet distribution is defined as

$$f(x;\alpha) = \frac{1}{B(\alpha)} \prod_{i=1}^{K} x_i^{\alpha_i - 1} \tag{17}$$

where x is a K-dimensional vector representing the data proportion for each client, $B(\alpha)$ is a normalization constant, and α_i is the control parameter. For each category, we generate a vector x that follows the Dirichlet distribution to determine the number of samples each client holds. By adjusting the values of α_i, we can control the uniformity of data distribution among clients. For instance, smaller α_i values result in data being concentrated in fewer clients, creating a highly imbalanced distribution. Conversely, larger α_i values lead to a more even distribution across all clients. This parameterization allows for flexible modeling of various real-world data distribution scenarios, enabling us to test the robustness of FL algorithms under different conditions.

In addition to the Dirichlet distribution, we also used a skewed distribution to generate heterogeneous data. The skewed distribution introduces asymmetry into the data, reflecting real-world scenarios where data distribution is often

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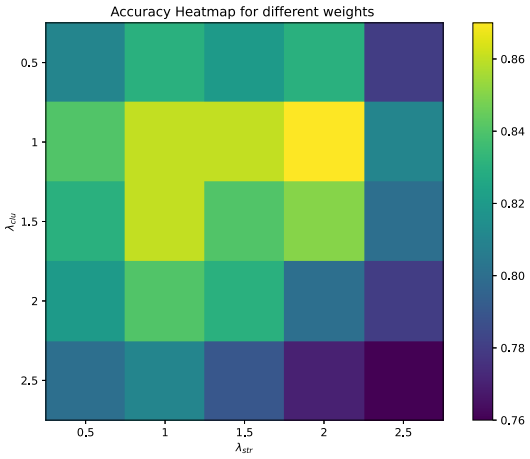


Fig. 11. Heatmap of accuracy under different loss weights.

imbalanced. Specifically, we sampled data for each category and introduced a skewness coefficient γ to control the degree of skewness. The skewness γ is defined as

$$\gamma = \frac{E\left[(X-\mu)^3\right]}{\sigma^3} \tag{18}$$

where X is the dataset, μ is the mean, and σ is the standard deviation. By adjusting the value of γ, we can control how the data is distributed among clients. Higher skewness values indicate that the data is more concentrated in fewer clients, while lower skewness values indicate a more even distribution across clients. To generate data with a skewed distribution, we first calculate the mean and standard deviation for each category and then adjust the data distribution by introducing the skewness coefficient, which causes the data to exhibit a certain degree of skewness among clients. This method effectively simulates the common real-world scenario of uneven data distribution.

By using these two different methods of data distribution, we are able to generate FL datasets with heterogeneous data distributions. This allows us to better simulate real-world data distribution scenarios and evaluate the performance and robustness of FL models under different data distribution conditions. The flexibility in controlling parameters such as $\boldsymbol{\alpha}$ for the Dirichlet distribution and γ for the skewed distribution enables comprehensive testing and optimization of FL algorithms. For the FAIR1M RS dataset, the data distribution generated by different methods is shown in Fig. 6.

In our experimental setup, non-IID data partitioning is applied as follows. Each sample has a unique index. We first iterate over the dataset by class, and in each iteration, we use a Dirichlet distribution (with α controlling the degree of heterogeneity) to randomly generate the proportion of the current class assigned to each client. Based on these proportions, we partition the dataset indices and distribute them among clients. After completing all iterations, each client stores its assigned data indices. The skewed distribution follows a similar approach but offers more precise control through the parameter γ. Specifically, we first iterate over the dataset by class and use γ to split each class into two subsets: a skewed portion and a uniformly distributed portion. We then determine the amount of skewed and uniform data assigned to each client and merge these two parts to construct the final dataset distribution for each client. The specific results of dataset partitioning are shown in Fig. 6. We used PyTorch 1.13 as the deep learning framework. The experiments were conducted on an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. We used stochastic gradient descent (SGD) as the optimizer, with a weight decay set to 0.0005. Choosing ResNet18 [60] as the backbone feature extraction network. We conduct 200 global communication rounds, with each client executing five local training epochs in every round. The initial learning rate for the model was set to 0.001.

C. Evaluation Method

We adopt two evaluation metrics, including classification accuracy and confusion matrix.

1) Classification Accuracy: Classification accuracy is a fundamental metric for evaluating the performance of a classification model. It measures the proportion of correctly classified samples out of the total samples. The classifier used in this work follows the standard setting adopted in [61], [62], and [63]: a single linear layer that maps the representation dimension directly to the label dimension. During evaluation, the learned representation vectors are kept frozen and only this linear layer is trained, taking an input whose size equals the representation dimensionality and producing an output that matches the number of target classes. We report accuracy under two evaluation settings: linear evaluation and semi-supervised learning. Linear evaluation follows the protocol in [61], [62], and [63]: the encoder is first trained entirely on the unlabeled dataset (encoder only, no classifier). After this unsupervised training, the encoder is frozen and a linear classifier is attached; the classifier is then trained for 100 epochs on the labeled training set, and the frozen encoder together with the trained classifier is evaluated on the test set to obtain the linear-evaluation accuracy (see Table I). Semi-supervised learning follows the procedure as in [61], [64], and [65], except that the linear classifier is trained with only a subset of the labeled data (e.g., 1%, 10%, and 50%). The frozen encoder and the trained head are finally evaluated on the test set, and the resulting accuracy is reported as the semi-supervised score (see Table II). The calculation formula is as follows:

$$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}\ +\text{FN}} \tag{19}$$

where true positive (TP) is the number of correctly classified positive samples, true negative (TN) is correctly classified negative samples, false positive (FP) is negative samples incorrectly classified as positive, and false negative (FN) is positive samples incorrectly classified as negative.

2) Confusion Matrix: The confusion matrix is a more detailed metric that provides insight into the performance of a classification model by showing the breakdown of correct and incorrect predictions for each class. It is a table with rows representing the actual classes and columns representing the predicted classes. The confusion matrix helps in identifying specific types of errors made by the model, such as confusion

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TABLE I
ACCURACY COMPARISON OF METHODS UNDER LINEAR EVALUATION PROTOCOL ON DIFFERENT DATASETS

Method	FAIR1M(%)			AID(%)			UCM(%)			NWPU(%)		
	IID	($\alpha = 0.01$)	($\gamma = 0$)	IID	($\alpha = 0.01$)	($\gamma = 0$)	IID	($\alpha = 0.01$)	($\gamma = 0$)	IID	($\alpha = 0.01$)	($\gamma = 0$)
FedSimCLR[64]	81.61	72.80	74.37	69.75	65.88	66.05	51.67	48.92	49.05	65.12	60.34	61.89
FedSimSiam[66]	72.23	67.72	69.21	56.25	55.39	53.21	49.56	47.24	47.89	60.45	57.12	58.03
FedBYOL[63]	80.97	73.28	74.14	57.50	58.53	56.33	52.67	50.19	49.41	62.87	59.98	60.45
FedMAE[67]	79.55	71.64	74.66	70.65	69.06	65.35	54.01	51.88	53.12	66.34	63.42	64.18
FedEMA[61]	85.37	78.28	79.64	75.69	74.78	71.69	58.94	56.73	57.15	70.29	67.54	68.47
FedUSC[68]	82.83	74.69	77.48	73.57	71.60	71.16	57.20	55.11	55.84	68.12	65.88	66.91
Fed-QSSL[69]	84.37	76.31	78.75	74.61	71.97	73.09	58.45	55.88	56.62	69.32	66.71	67.55
FeatARC[70]	86.50	80.35	81.22	76.75	73.26	74.83	60.10	58.02	59.10	71.45	69.02	70.21
DURL-PU	**87.54**	**82.36**	**82.39**	**77.20**	**76.01**	**78.73**	**61.50**	**60.18**	**61.24**	**73.68**	**70.89**	**71.75**

TABLE II
ACCURACY COMPARISON ON USING 1%, 10%, AND 50% OF LABELED DATA FOR SEMI-SUPERVISED LEARNING ON NON-IID SETTINGS OF DIFFERENT DATASETS

Method	FAIR1M(%)			AID(%)		
	1%	10%	50%	1%	10%	50%
FedSimCLR[64]	59.49	70.80	71.55	60.50	64.37	65.87
FedSimSiam[66]	56.17	63.32	65.64	42.07	50.89	52.27
FedBYOL[63]	60.37	69.94	71.19	50.53	55.48	56.11
FedMAE[67]	54.12	69.18	75.36	57.42	61.46	63.30
FedEMA[61]	63.49	72.25	77.64	62.76	67.07	68.82
FedUSC[68]	60.24	70.69	73.21	59.63	64.09	64.79
Fed-QSSL[69]	61.33	73.58	75.30	63.86	66.98	65.47
FeatARC[70]	56.50	75.39	78.18	64.66	69.64	72.05
DURL-PU	**71.25**	**79.95**	**80.86**	**70.43**	**74.42**	**76.28**

between certain classes, and provides a basis for further analysis and improvement of the classification algorithm.

D. Object Classification Experiments

The experiment consists of three main parts. The first part compares the proposed algorithm with other advanced methods across different datasets. The second part evaluates the performance under different non-IID data settings to verify the effectiveness of the proposed algorithm in handling heterogeneous data distributions. The third part involves ablation studies to verify the effectiveness of important hyperparameters in the proposed method.

1) Experimental Comparison With Other Methods on Different Datasets: Various unsupervised representation learning algorithms are selected for experimental comparison, such as SimCLR [64], SimSiam [66], BYOL [63], MAE [67], as well as the FedEMA [61] and FeatARC [70] algorithms. SimCLR achieves representation learning through contrastive learning on different data augmentations of images. SimSiam, without using negative samples, prevents feature collapse of different views by employing a stop-gradient technique. Similarly, BYOL does not require negative samples and uses a cross correlation loss function to align the features of different views of the same image as closely as possible. MAE is a generative-based representation learning method that masks a high proportion of the input image and recovers the original image at the network's output layer. This method, based on the transformer network structure, cleverly designs an effective self-supervised task, resulting in excellent generalization performance. To address the issue of data heterogeneity, FedEMA adopts divergence-aware predictor update (DAPU) to determine model update strategies and introduces the Federated Exponential Moving Average update, which dynamically balances partial local knowledge and the global model based on the divergence scale. FeatARC groups clients according to the distribution of their data and introduces an auxiliary loss function to align the local model with the global model. However, existing algorithms do not take into account the consistency of model parameter distributions across different clients, which leads to performance degradation under data heterogeneity. I believe our algorithm can effectively mitigate this issue and offer a promising solution to improve distribution consistency in existing methods.

We evaluate the performance of the proposed DURL-PU framework and other methods on the FAIR1M, AID, and UCM datasets, with data divided into IID and non-IID distributions. The non-IID data was sampled using a Dirichlet distribution with $\alpha = 0.01$ and a skew distribution $\gamma = 0$. The accuracy results for the proposed method and other methods are reported in Table I. When α is set to 0.01, the data exhibits a Dirichlet heterogeneous distribution. When γ is set to 0, the client data shows a skewed distribution, meaning that in the FAIR1M dataset, each client's data comes from four different classes. The specific data heterogeneity distribution is shown in Fig. 6.

From the overall results, it is evident that the accuracy on the FAIR1M dataset is significantly higher than that on the AID and UCM datasets under different data settings. This is because unsupervised representation learning requires a large amount of unlabeled training data to achieve good performance, and both the AID and UCM datasets are much smaller in size compared to FAIR1M, which accounts for this accuracy difference. The limited sample size in AID and UCM restricts the diversity and richness of data, making it challenging for unsupervised methods to fully capture meaningful representations. Under the Dirichlet non-IID setting in the FAIR1M dataset, the proposed DURL-PU method achieved the top-1 accuracy of 82.36%, outperforming all other methods, including FedSimCLR, FedSimSiam, FedBYOL, FedMAE, FedEMA, FeatARC, FedUSC, and Fed-QSSL by margins of {9.56%, 14.64%, 9.08%, 10.72%, 4.08%, 2.01%, 7.67%, and 6.05% respectively}. Similarly, in the same non-IID setting on the AID dataset, DURL-PU achieved the highest accuracy

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of 76.01%, surpassing FedSimCLR, FedSimSiam, FedBYOL, FedMAE, FedEMA, FeatARC, FedUSC, and Fed-QSSL by {10.13%, 19.76%, 17.48%, 6.95%, 4.08%, 2.01%, 4.41%, 4.04% respectively}. On the UCM dataset under the same Dirichlet non-IID setting, DURL-PU achieved a top-1 accuracy of 60.18%, outperforming FedSimCLR, FedSimSiam, FedBYOL, FedMAE, FedEMA, FeatARC, FedUSC, and Fed-QSSL by margins of {11.26%, 12.94%, 9.99%, 8.30%, 3.45%, 2.16%, 5.07%, 4.30% respectively}. On the NWPU dataset under the same Dirichlet non-IID setting, DURL-PU achieved a top-1 accuracy of 70.89%, outperforming FedSimCLR, FedSimSiam, FedBYOL, FedMAE, FedEMA, FeatARC, FedUSC, and Fed-QSSL by margins of {10.44%, 11.92%, 9.15%, 7.82%, 3.21%, 2.04%, 5.68%, 4.97% respectively}. These results demonstrate that DURL-PU consistently achieves superior performance across datasets, effectively addressing data heterogeneity in FL.

Scene image classification is characterized by small inter-class separations and large intra-class variations, often leading to classification confusion. Our proposed method addresses this challenge by incorporating high-dimensional semantic information, enhancing intra-class compactness, and increasing inter-class separability. This approach effectively mitigates the classification confusion caused by these characteristics.

To further analyze the classification details, we specifically plotted the confusion matrices of various methods, as shown in Fig. 7. The confusion matrices represent different categories from left to right, such as aircraft and ships. The visualized results reveal that the subcategories of aircraft (Boeing 747 and A321) and ships (dry cargo ship and liquid cargo ship) pose significant challenges in RS image classification. In the cases of FedSimCLR, FedSimSiam, FedBYOL, and FedMAE, there is a substantial amount of misclassification, mainly concentrated in the Boeing 747 and A321 categories. This is due to the very similar visual features of the two aircraft, which lead to easy misidentification. Other methods also commonly misidentify ships. The proposed DURL-PU method achieves better recognition results in most categories. Although the confusion matrix reveals that DURL-PU achieves superior performance across most categories, however, its performance for the "Engineering Ship" category highlights a notable limitation. Specifically, DURL-PU exhibits confusion with "Tugboat," attaining an accuracy of only 70%, which is 5% lower than the baseline FedSimCLR, 2% lower than FedMAE, and 1% lower than both FedUSC and FedBYOL. This limitation may stem from the additional clustering process in DURL-PU. While it effectively enlarges decision boundaries between different classes, it might occasionally cause confusion in distinguishing highly similar and visually fine-grained targets.

Additionally, we compared the performance of the algorithms under semi-supervised learning. The overall accuracy (OA) of the algorithms is shown in Table II. The experimental results indicate that, across different datasets and label data ratio settings, our method consistently outperforms other methods. Notably, with only 1% labeled data, our method demonstrates a significant advantage over the state-of-the-art FeatARC, which further proves the effectiveness of our approach.

TABLE III
ACCURACY COMPARISON ON USING DIFFERENT BACKBONE NEURAL NETWORK ON NON-IID SETTINGS OF FAIR1M DATASET

Method	ResNet18		VGG16	
	IID	Non-IID	IID	Non-IID
FedSimCLR[64]	81.61	72.80	75.52	66.95
FedSimSiam[66]	72.23	67.72	68.01	61.41
FedBYOL[63]	80.97	73.28	72.36	64.23
FedMAE[67]	79.55	71.64	71.71	63.04
FedEMA[61]	85.37	78.28	78.84	69.19
FedUSC[68]	82.83	74.69	75.31	67.58
Fed-QSSL[69]	84.37	76.31	75.97	69.74
FeatARC[70]	86.50	80.53	77.49	69.85
DURL-PU	**87.54**	**82.36**	**81.74**	**75.48**

TABLE IV
ACCURACY COMPARISON ON VARIOUS NUMBERS OF CLIENTS ON THE NON-IID SETTING OF FAIR1M DATASET

Method	Number of device		
	5	10	25
FedSimCLR[64]	72.80	70.80	64.30
FedSimSiam[66]	67.72	64.12	61.94
FedBYOL[63]	73.28	69.47	65.25
FedMAE[67]	71.64	68.07	65.31
FedEMA[61]	78.28	75.33	72.25
FedUSC[68]	72.31	71.97	69.37
Fed-QSSL[69]	76.09	72.61	69.91
FeatARC[70]	80.35	76.91	70.87
DURL-PU	**82.36**	**79.34**	**78.50**

2) Impact of Different Parameter Settings on Algorithm Performance: We conduct experimental analysis using different parameter settings. We first selected different backbone networks for feature extraction, specifically ResNet18 and VGG16. As shown in Table III, the experimental results demonstrate that ResNet18 consistently achieved better results across various data distributions, which were used for our proposed method.

To validate the scalability of the proposed method, we evaluated it with 5, 10, and 25 clients on the FAIR1M dataset with the Dirichlet index set to 0.01. We report the accuracy of the proposed method and other methods in Table IV. The experimental results indicate that the proposed method demonstrates robustness and superior accuracy compared to other methods across different numbers of clients. As shown in Table IV, the performance of all approaches begins to decline as the number of clients increases. However, the accuracy degradation of the proposed method is significantly smaller than that of other methods. The accuracy of DURL-PU decreased by only 3.86% when the number of clients increased from 5 to 25, while FeatARC dropped by 9.84%. This indicates that the proposed method demonstrates better robustness across different scales of edge clients.

We also evaluated the impact of different local training parameters on model performance, as shown in Fig. 8.

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DURL-IN refers to training the representations using only the instance-level loss $\mathcal{L}_{\text{in}}$. We conducted these experiments on the FAIR1M dataset under a non-IID setting ($\alpha = 0.01$). Fig. 8(a) illustrates the effect of various local training epochs on model performance. We found that the performance with epoch = 5 and epoch = 10 is very close and superior to other epoch settings, indicating that the number of local training epochs should neither be too large nor too small. Besides, our DURL-PU outperforms FeatARC and FedSimCLR across all epoch settings. Fig. 8(b) shows that the model achieves the best performance when the local training batch size is set to 256. These results indicate that the setting of certain client parameters is crucial for the final performance of the algorithm.

Fig. 9 illustrates the accuracy of the FAIR1M dataset under different data heterogeneity settings. In this experiment, we can observe that as the heterogeneity of client data distribution becomes more extreme, the model's accuracy becomes progressively lower. However, our method consistently outperforms other approaches across various levels of data heterogeneity.

To assess the quality of the learned representations, we feed the test samples into the trained models and utilize the t-SNE technique to visualize the extracted features. We visualize the features from the penultimate layer, which is the last layer before the classification head. These visualizations illustrate and compare the feature distributions learned by different methods, as shown in Fig. 10. It can be observed that DURL-PU significantly improves feature representation by not only separating samples from different categories more distinctly but also achieving a more uniform distribution of features. This demonstrates that DURL-PU enhances the separability of class representations while ensuring better global uniformity, effectively addressing the challenges posed by data heterogeneity in FL. Additionally, we use stars to mark the prototypes of different scene classes. It can be observed that the prototypes of the proposed DURL-PU method are more uniformly distributed compared to other methods, which strongly demonstrates the effectiveness of the proposed method.

Fig. 11 illustrates the impact of different loss weight combinations on model accuracy. It is evident that when the weight of λ_{str} is set to 2, the accuracy reaches its highest value across all configurations. This highlights the importance of balancing the structural loss with an optimal weight to achieve superior performance. In FL, enhancing the consistency of weights across different clients is crucial for addressing data heterogeneity, which makes it necessary for λ_{str} to have a relatively larger weight to effectively improve model performance under non-IID settings.

To evaluate the impact of cross-dataset heterogeneity on FL, we conducted experiments where each client was trained on a different dataset: FAIR1M, AID, UCM, and NWPU. As shown in Fig. 12, we compared the performance of several FL algorithms, including FedEMA, FedUSC, Fed-QSSL, FeatARC, and our proposed method, DURL-PU. The results reveal significant performance variations across datasets, underscoring the effect of cross-dataset heterogeneity on FL. Notably, DURL-PU consistently demonstrates strong competitiveness

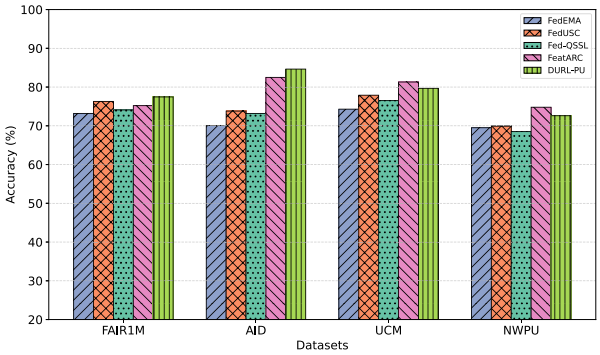


Fig. 12. Impact of cross-dataset heterogeneity on FL performance.

TABLE V
COMPARISON OF MODEL PARAMETERS AND GFLOPS WITH DIFFERENT METHODS ON THE NON-IID SETTING OF THE FAIR1M DATASET

Method	Architecture	Params	GFLPOS	OA(%)
FedSimCLR[64]	ResNet18	11.70M	3.65	72.80
FedSimSiam[66]	ResNet18	12.03M	3.65	67.72
FedBYOL[63]	ResNet18	22.94M	7.28	73.28
FedMAE[67]	ResNet18	39.97M	7.39	71.64
FedEMA[61]	ResNet18	38.47M	7.40	78.28
FedUSC[68]	ResNet18	28.34M	7.25	72.31
Fed-QSSL[69]	ResNet18	38.63M	5.72	76.09
FeatARC[70]	ResNet18	16.61M	6.97	80.53
DURL-PU-Light	MobileNetv3	**9.18M**	**0.75**	68.74
DURL-PU	ResNet18	18.85M	5.47	**82.36**

across all datasets, achieving the highest accuracy on AID. These findings indicate that despite substantial heterogeneity among datasets, DURL-PU effectively adapts to diverse data distributions while maintaining superior model performance.

3) Analysis of the Model Parameter and Computational Complexity: We conducted an analysis of the model parameter counts and computational complexity (GFLOPs) for various methods, as detailed in Table V. DURL-PU-Light represents a lightweight variant of the proposed DURL-PU framework, in which the backbone network is replaced with MobileNetV3 to reduce model complexity. Traditional methods such as FedSimCLR and FedSimSiam exhibit relatively low parameter counts (11.70M and 12.03M, respectively) and moderate GFLOPs (3.65), but their OA is constrained under the nn-IID settings, indicating limitations in their ability to generalize across heterogeneous data distributions. Advanced methods like FedBYOL, FedEMA, and Fed-QSSL improve accuracy but at the cost of significantly increased parameter counts (22.94M–38.63M) and computational complexity (5.72–7.40 GFLOPs), which could pose challenges in resource-limited FL environments. Although DURL-PU achieves the highest accuracy across various datasets, its model parameters and GFLOPs are 37% and 33% higher compared to the baseline FedSimCLR. The additional computational cost is primarily attributed to the clustering loss computation introduced in DURL-PU, which enhances feature separation but at the expense of greater complexity.

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TABLE VI
ABLATION STUDIES OF DIFFERENT COMPONENTS ON FAIR1M DATASET UNDER NON-IID SETTING

Model	ResNet18(In)	Clu	Str	OA(%)
1	√			71.85
2	√	√		74.51
3	√		√	78.96
4	√	√	√	82.36

TABLE VII
ABLATION STUDIES OF DIFFERENT COMPONENTS ON AID DATASET UNDER NON-IID SETTING

Model	ResNet18(In)	Clu	Str	OA(%)
1	√			69.75
2	√	√		72.33
3	√		√	75.09
4	√	√	√	77.20

In contrast, our proposed method, DURL-PU-Light, achieves a notable reduction in both parameter count and GFLOPs, with just 9.18M parameters and 0.75 GFLOPs, while maintaining competitive performance with an OA of 68.74%. This highlights its efficiency and suitability for resource-constrained distributed systems. Additionally, DURL-PU strikes a balance between accuracy and computational cost, achieving the highest OA of 82.36% with a moderate parameter count of 18.85M and 5.47 GFLOPs. These results demonstrate the superior trade-off offered by our proposed methods compared to existing approaches, showcasing their potential for scalable and efficient FL in real-world applications.

4) Ablation Study for Clu and Str: We conducted a series of experiments to verify the effectiveness of different components of our proposed DURL-PU method. We used the FAIR1M and AID datasets under non-IID settings for these experiments. ResNet18(In) represents the backbone network ResNet18 with an instance loss, while Clu and Str represent the cluster loss and structure loss, respectively. The experimental results are shown in Tables VI and VII. The results demonstrate that the proposed components significantly improve the model's performance, especially the Str component, which achieves an accuracy improvement of more than 5% on both datasets.

In addition, we carefully selected representative challenging RS samples for visualization analysis, including subcategories of airplanes and ships that are prone to misidentification. We used gradient-weighted class activation mapping (Grad-CAM) [71] to visualize the neural network's feature maps, which intuitively reflect the network's regions of interest. The visualization results are shown in Fig. 13. As we incorporated our components, the model progressively began to focus on key, distinguishable areas of the images, avoiding irrelevant background regions, and covering the target objects more comprehensively. These feature map analysis results further demonstrate the effectiveness of the proposed components.

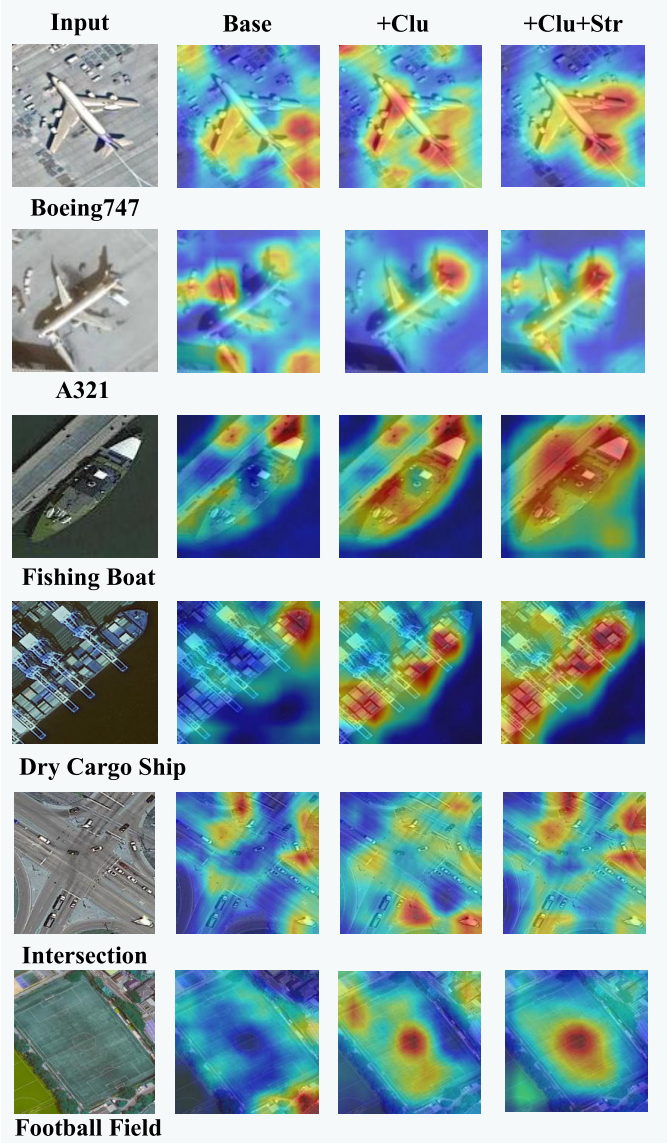


Fig. 13. Grad-CAM feature map visualization results.

V. CONCLUSION

In this work, we propose a distributed unsupervised representation learning neural network, DURL-PU, to address the problem of heterogeneous data distribution and the lack of high-quality labeled data across different satellites. The DURL-PU explores the instance-, cluster-, and structure-level representation to progressively enhance the uniformity, which can mitigate dimensional collapse caused by data heterogeneity. We conducted extensive experiments and analyses on the FAIR1M, AID, UCM, and NWPU datasets to validate the effectiveness of the proposed method. The results demonstrate that our method outperforms current mainstream methods across various experimental scenarios, including different heterogeneous data settings, label ratios, and client scales. However, the proposed method has only been applied to the basic task of RS object classification. In future work, the potential of distributed unsupervised representation learning in the RS field should be further explored. We aim to extend the learned representations beyond object classification to more

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universal and critical downstream RS tasks, such as object detection, change detection, and semantic segmentation.

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